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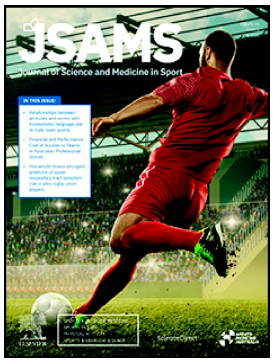

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Calf muscle injury in elite athletics: an 8-season prospective cohort study

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Abstract

Objective: To describe epidemiological data of indirect calf muscle injury (CMI) in elite track and field athletes. To investigate relationships between CMI and athlete characteristics to inform future injury prevention strategies**.**

Design: Descriptive epidemiology study

Methods: A total of 201 track and field athletes, mean age 25.9 +/- 4.4, were prospectively observed over 8-seasons (616 athlete seasons). The main outcome measures were injury location, injury number, incidence, incidence proportion, severity, burden and recurrence. Poisson generalised linear regression models were used to explore relationships between potential risk factors.

Results: There were 85 CMIs during the study period. Thirty three percent of CMIs were classified as severe. Soleus injuries had the highest incidence, burden and recurrence and were present across all event groups. Incidence proportion of all CMIs was 11.1%, with the highest incidence of injuries occurring in April. Acute CMI was the most common mechanism, with the right lower limb (58%) being more affected than the left (42%). There was no association between CMI occurrence and sex, though there was a significant positive effect between both age and previous lower limb injury.

Conclusion:

Calf muscle injury is prevalent in elite track and field athletes. CMI has a peak incidence in April which can have a detrimental effect on an athlete's performance goals for the impending summer competition season. The soleus muscle is the most injured muscle with the biggest burden and recurrence making it a muscle of focus when developing rehabilitation and prevention programs. Older athletes, and athletes with a previous lower limb injury, are more at risk of CMI in elite athletics. Due to the burden and potential performance impact of CMI, further injury prevention efforts are required to reduce the impact of this injury type.

Key Words: Athletic injury, Epidemiology, Athletics, Muscles, Triceps Surae, Risk Factors

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Introduction

Calf muscle injury (CMI) is one of the most common injuries in running athletes with high levels of incidence and prevalence in youth, recreational, elite and older populations.[1–3] Within elite track and field athletics it is the second most common soft tissue injury and has a similar incidence to other running based sports such as Australian Rules Football (AFL), soccer and rugby union.[4–7] Muscle injuries in elite athletics are very common and represent the most burdensome type of injury in a competition setting, with the hamstring muscles specifically having the highest incidence.[8,9] Though there has been much research and acknowledgement of the burden of hamstring injury in the literature, less is known epidemiologically about CMIs in elite athletics cohorts [7–10].

Athletics has maximal physiological and musculoskeletal demands within training and competition.[11,12] Task-specific demands of athletics include running and jumping, with most athletics events involving running, whether that be acceleration, maximal velocity sprinting, steady state running, curvilinear running, or run-up approaches during field events.[11–13] The soleus and gastrocnemius are important muscular power contributors across all running modalities.[14,15] The soleus muscle specifically has been estimated to produce forces of approximately eight times body weight during running speeds of up to 7 m/s, and is the primary contributor to impulse forces at these running speeds [14,15]. Furthermore, in modelling studies, ankle plantar flexors have also been demonstrated to produce the greatest amount of positive work during acceleration.[16] It is these known physiological demands of the calf musculature, in combination with the key activity demands of running and jumping in athletics, that put athletics participants at high risk of CMI.[7,15,16] Research can help understand how this injury affects athletes in their training and competition preparation.

Little epidemiological research has been published focusing on CMI in elite athletics.[7-10] During elite athletics competitions it has been observed that lower leg muscle strain is the second most common injury location compared to the posterior thigh.[8] Lower leg muscle injuries tend to occur more frequently in endurance running athletes who run at lower velocities compared to power-based athletes (sprinters and jumpers), however, the specific lower leg muscles injured within a competition

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setting is unknown.[9] Outside of elite athletics competition there is only one study that has reported CMI epidemiology yet there were only a small number of cases, therefore, CMI can occur within elite athletics though more detailed analysis is needed.[7]

The primary aim of this research was to describe the epidemiological characteristics of indirect CMI in elite track and field athletes with respect to the muscle group, location and the incidence, incidence proportion, recurrence, severity and recurrence of each specific injury. The secondary aim of this study was to investigate potential associated characteristics with this injury presentation which may help inform injury prevention practice.

Methods

Study Design

A prospective observational study design of Olympic athletes on the British Athletics World Class Programme (WCP) was employed with a data collection period spanning from the 1st of Dec 2011 until the 30th of November 2019. The British Athletics WCP season starts each funded season on the 1st of December and runs to the 30th of November the following year, data collection, therefore, spanned 8 full seasons. Ethical approval for the study was granted by the University College London Ethics Committee (5361/002) in accordance with the Declaration of Helsinki. The STROBE Extension for Sport Injury and Illness Surveillance was used to guide the reporting of this study.[17]

Participants and Setting

A total of 201 athletes (males: 109 athletes, 331 athlete-seasons, mean age – 24.4; females: 92 athletes, 285 athlete-seasons, mean age – 25.1) accepted on the British Athletics WCP were included in this study. Inclusion criteria for this study was selection onto the programme in any given season. Selection onto the program was reviewed annually. There were no exclusion criteria and athletes were lost to the study if they were exited from the WCP for performance reasons. The average time that any athlete was part of the program during the study period was 3.1 +/- 2.0 seasons (range: 1 – 8 seasons). Events represented in the program were: *sprints*/*sprint relays* (60 m, 100 m , 200 m, 400 m, 4 ×100 m

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and 4 × 400 m); *middle distance* (800 m and 1500m), *long-distance* (3000 m, 5000m, 10000m and 3000 m steeplechase), *marathon, race walking* (20 km, 50 km), hurdles (60 m, 100 m, 110 m, 400 m), *jumps* (high jump, long jump, triple jump, pole vault), *throws* (discus, javelin, hammer, shot put) and *combined events* (decathlon, heptathlon).[18] Athletes were entered onto the program if they were considered a genuine prospect for a medal at a global championship. Athletes on the program were predominantly based in the UK, however, some athletes trained overseas either as their permanent residence or during camps. Injury data were collected across all training and competition venues used during the study period.

Data collection procedures

Injury Data

Suspected indirect CMIs were initially assessed via a detailed clinical examination by an experienced Chartered Physiotherapist and a Sport & Exercise Medicine Doctor. A clinical diagnosis of CMI was made if the athlete reported an indirect mechanism and had pain on stretch, pain on palpation, pain on contraction and any/or loss of calf muscle function. Where possible, athletes were referred for radiological investigation to confirm the diagnosis. Clinical assessment for CMI would normally take place within 48hrs of the injury and would be performed at the athletes training base. All CMIs were recorded on a cloud-based athlete management system Teamworks AMS (formerly Smartabase, North Carolina, USA).[19] BAMIC, or any other, classification was not used for all injuries as data collection pre-dated the implementation of this muscle grading system. Following the conclusion of data collection, injuries were downloaded from the athlete management system and filtered to include only indirect CMIs.

Injury Definitions

Medical attention injuries were defined as: *any musculoskeletal complaint that required medical attention from a physiotherapist*. [18] Time-loss injuries were defined as: *any injury that prevented an athlete from taking a full part in training or competition that would normally be required of them for a period equal to or greater than 24 hours after the injury was sustained.* [18] Time-loss duration was

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calculated as the number of days for which the athlete was unable to fully participate in event-specific training because of the injury. The athlete's return to full training followed a shared decision between the athlete, coach and practitioner, based on symptomology, physical capacity, psychological readiness, and previously completed event-specific training. Injury severity was classified as either *no time-loss* (0 days), *mild* (1–7 days), *moderate* (8–28 days), or *severe* (> 28 days) based on the number of days of time-loss.[11] Muscle injury recurrence was defined as a subsequent injury to the same location and same tissue once the athlete had fully returned to training or competition.[17,20] Recurrence time frames were limited to 12 months post index injury.[20] Exacerbations were defined as a worsening of the original injury before an athlete had fully returned to training or performance.[17]

For athletes within the United Kingdom, injury data were recorded for all CMI presentations as part of their usual care. For athletes based overseas, contact was made weekly by the medical team to capture data on training status, injury and illness. Following each medical appointment, the athlete's training status was categorised as: *unable to train*, *able to train with activity modification,* and *fully training*. Medical personnel reviewed injury records weekly to confirm training status was current.

Injury mode of onset was classified based on the mechanism (i.e., *repetitive* or *acute*) and the presentation (i.e., *gradual onset* or *sudden onset*).[17] Subject characteristics were recorded including sex, age, and event group as classified by Timpka et al (2014)[18]. Medical personnel were trained during their departmental onboarding on injury surveillance definitions and procedures as part of standard medical induction.

Data Analysis

Following the completion of the data collection period, all data were downloaded from the athlete management system for analysis, following which two authors (SK and NP) reviewed all CMI records for clinical accuracy. The proportion of injuries by mode of onset, occurrence, severity scale, muscle, and side; mean injury severity by muscle, event group, and occurrence; and injury burden by event group and muscle; were calculated with bootstrapped 95% confidence intervals. Each season's incidence proportion was calculated for all participants, and by event group, where the denominator

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was the count of athletes in the British Athletics WCP each season, and the numerator was the count of athletes injured during the season.

Statistical Analysis

Poisson generalised linear regression models were used to model medical attention and time-loss CMI incidence across the eight seasons. The injury count was included as the dependent variable, whilst athlete sex, event group, and age were included as independent variables. Estimated marginal mean incidence rates were extracted and back-transformed using the *emmeans* R package to calculate the incidence rate per athlete-season. Significance was set at $p \leq .050$. All data and statistical analysis took place in R v.4.4.1 (R Foundation for Statistical Computing, Vienna, Austria). A post-hoc power calculation was performed, a simulation of 1000 experiments revealed moderate power to detect a smallest effect of interest for age (rate ratio - 1.05, 57.9%); previous lower limb injury (rate ratio - 1.2; 66.1%); and a low power for sex (rate ratio 1.2; 18.2%). Subsequently, some results should be interpreted with caution.

Results

Across the eight seasons of data collection, 616 athlete-seasons were recorded, during which 85 CMIs were observed, 77 of which resulted in time-loss. Fifty-two injuries affected the soleus (61% [51 to 71%]), whilst 26 affected the gastrocnemius (31% [21 to 41%]), and 7 (8% [2 to 14%]) were not attributed to a specific muscle. Forty-nine injuries occurred to the right leg (58% [47 to 68%]) and 36 injuries occurred to the left leg (42% [32 to 53%]).

The count of injuries affecting the right and left legs, respectively, was the same in endurance (1 vs 1), hurdles (12 vs 12), and throws (1 vs 1) but different in jumps (7 vs 2), middle-distance (9 vs 5), multi-events (2 vs 4), and sprints (17 vs 11). Fifty-four injuries were classified as sudden-onset (64% [53 to 74%]), whilst 31 were classified as gradual-onset (37% [26 to 47%]). All injuries were classified as being directly related to training or competition. The count of injuries across the eight seasons, and across the year, is presented in Figure 1.

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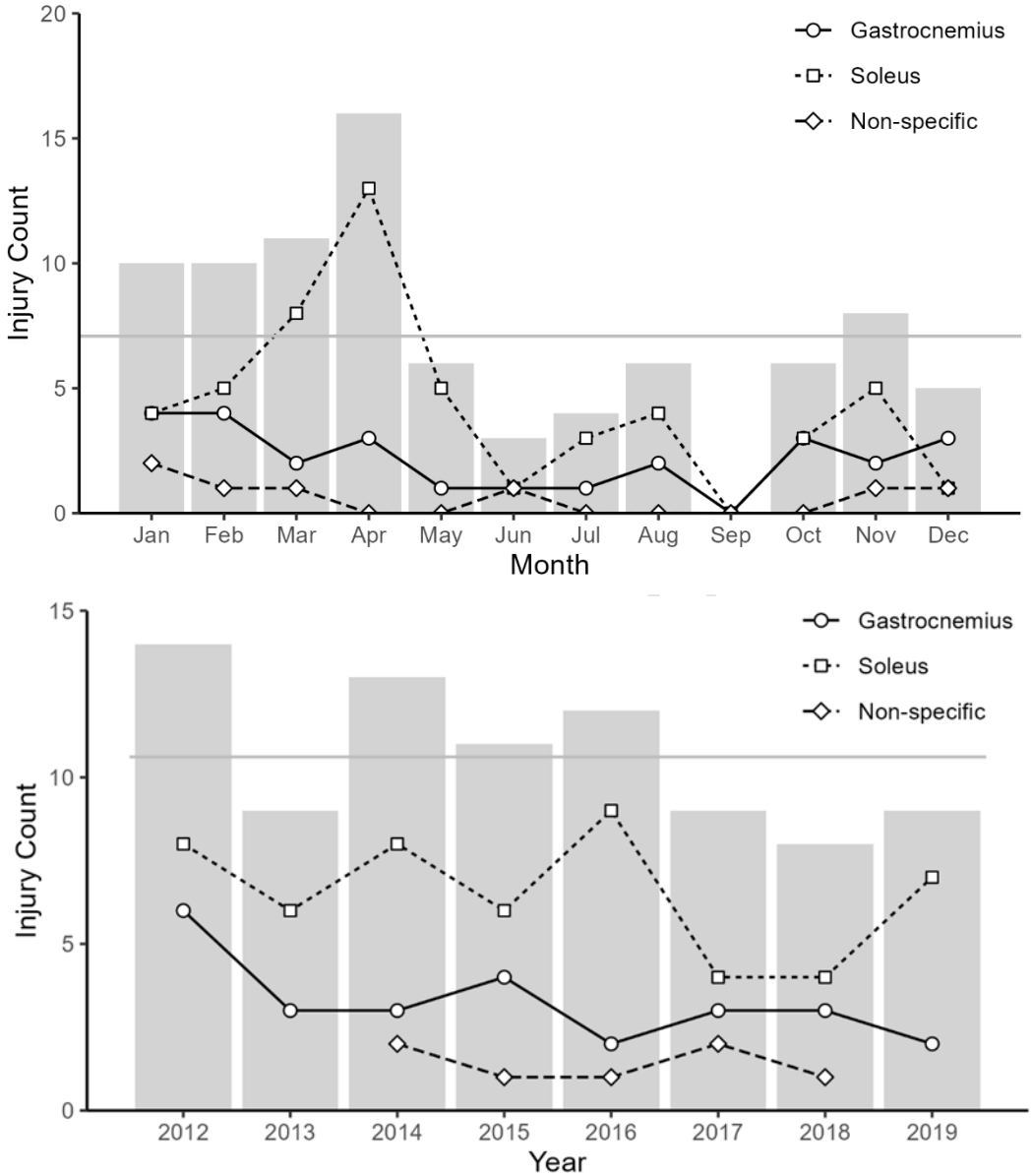

Figure 1 – Muscle injury count by month, and by season for the soleus, gastrocnemius, and unknown triceps surae muscles

Injury Incidence

Across all participants and event groups, the raw incidence of medical attention injuries was 0.14 [0.11 to 0.17] injuries·athlete season^{-1} (male: 0.16 [0.12 to 0.21] injuries·athlete season^{-1}; female: 0.12 [0.08 to 0.16] injuries·athlete season^{-1}). The raw incidence rate of time-loss injuries was 0.13 [0.10 to 0.16] injuries·athlete season^{-1} (male 0.15 [0.11 to 0.19];female: 0.10 [0.07 to 0.15] injuries·athlete season^{-1}). No significant effect of sex was observed on either medical attention (incidence rate ratio =

1.48, p = .200) or time-loss injuries (incidence rate ratio = 1.56, p = .162). Differences in injury incidence rates across event groups are presented in Figure 2. A significant positive effect of age was observed, with each additional year equating to a 1.09 [1.01 to 1.17] times greater medical attention incidence (p = .0025), and a 1.11 [1.03 to 1.19] times greater time-loss injury incidence (p = .005). A significant positive effect of previous lower limb injury count was observed on medical attention injuries, with each additional injury equating to a 1.21 [1.00 to 1.42] times greater medical attention incidence (p = .025; Time-loss incidence: 1.11 [0.94 to 1.38], p = .128).

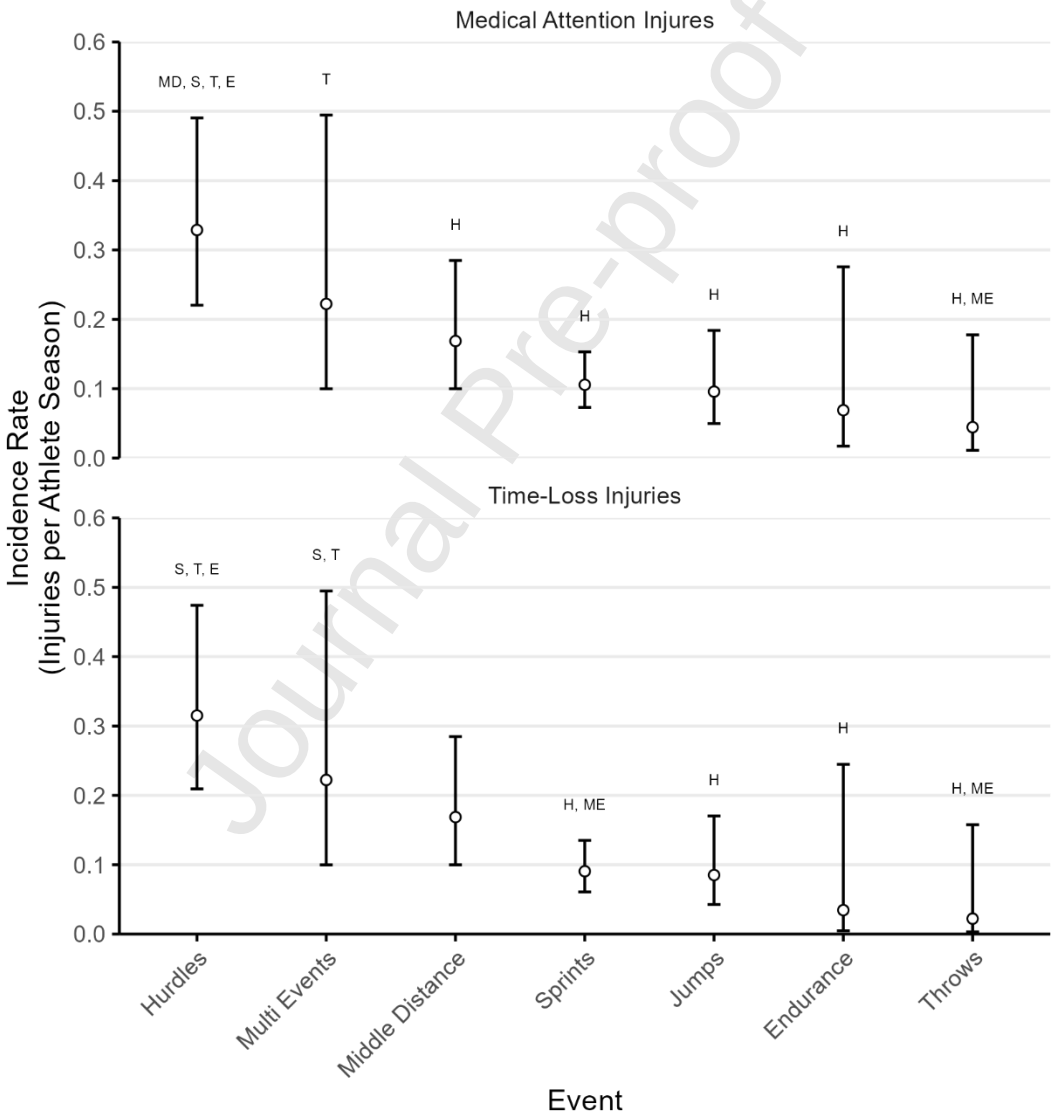

Figure 2 – Incidence rates of medical attention and time loss injuries across event groups. Letters denote events to which the event's incidence rate is significantly different: H – Hurdles; ME – Multi-Event; MD – Middle Distance; S – Sprints; J – Jumps; E – Endurance; T – Throws.

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Incidence Proportion

The mean single-season incidence proportion was 11.1%, and the median single-season incidence proportion was 10.2% (range 9.3% - 14.3%).

Injury Recurrence

Fourteen injuries were recurrences, whilst one injury was an exacerbation. The mean time-to-recurrence was 364 days, whilst the median time-to-recurrence was 214 days (range: 32 to 1418 days). The exacerbation occurred three days following the initial injury. Twenty-one injuries were preceded by at least one muscle injuries to the same calf. One-hundred-and-forty-nine participants incurred no calf injuries, 34 athletes had one injury, and 18 athletes had multiple calf injuries (two injuries – 13 athletes; three injuries – two athletes; four injuries – one athlete; five injuries – one athlete; 10 injuries – one athlete). Mean injury severity was similar for both the index injury (33.2 [24.9 to 42.7] days) and recurrences (29.7 [17.1 to 45.7] days).

Injury Severity and Time-Loss Burden

Of the 85 injuries, seven resulted in no time-loss (9% [4–16%]), 13 were *minor* (15% [8–23%]), 37 were *moderate* (43% [33–54%]), and 28 were *severe* (33% [23–43%]). The mean time-loss was 29 [23–37] days, and the median time-loss was 19 [16–24] days. Injury severity by event, and the time-loss burden of calf strain injuries across events is presented in Table 1.

Table 1 – CMI incidence, mean severity and time-loss burden by cohort and event group

	Athletes (n)	Athlete-Seasons (n)	Incidence (injuries per athlete-season)	Mean Severity (days of time-loss)	Time-loss Burden (days lost per athlete-season)
Overall					
All	**201**	**616**	**0.14 (0.11-0.17)**	**29.5 (22.5-38.0)**	**4.1 (2.9-5.6)**
Gastroc			0.04 (0.03-0.06)	27.1 (12.8-46.6)	1.14 (0.5-2.2)
Soleus			0.08 (0.06-0.11)	31.1 (23.6-38.8)	2.6 (1.7-3.6)
Non-specific			0.01 (0.01-0.02)	27.0 (11.0-53.7)	0.3 (0.1-0.7)
Endurance					
All	**14**	**29**	**0.07 (0.02-0.28)**	**8.0 (0.0-16.0)**	**0.6 (0.0-1.7)**
Gastroc			0.03 (0.00-0.24)	0.0 (0.0-0.0)	0.0 (0.0-0.0)

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Soleus			-	-	-
Non-specific			0.03 (0.00-0.24)	16	0.6 (0.0-1.7)
Hurdles					
All	**19**	**73**	**0.33 (0.22-0.49)**	**27.8 (18.3-38.9)**	**9.2 (4.1-15.3)**
Gastroc			0.11 (0.05-0.22)	19.9 (9.1-32.5)	2.2 (0.3-5.2)
Soleus			0.22 (0.13-0.36)	31.8 (19.2-46.9)	7.0 (2.7-12.2)
Non-specific			-	-	-
Jumps					
All	**29**	**94**	**0.10 (0.05-0.18)**	**21.7 (8.3-40.2)**	**2.1 (0.4-4.5)**
Gastroc			0.03 (0.01-0.10)	30.7 (0.0-86.0)	1.0 (0.0-2.9)
Soleus			0.05 (0.02-0.13)	18.4 (9.0-28.2)	1.0 (0.0-2.3)
Non-specific			0.01 (0.00-0.08)	11	0.1 (0.0-0.4)
Middle Distance					
All	**26**	**83**	**0.17 (0.10-0.28)**	**35.3 (21.6-50.9)**	**6.0 (2.4-10.4)**
Gastroc			0.01 (0.00-0.09)	16	0.2 (0.0-0.6)
Soleus			0.13 (0.07-0.24)	31.9 (18.9-45.6)	4.2 (1.4-8.0)
Non-specific			0.02 (0.01-0.10)	63.5 (24.0-103.0)	1.5 (0.0-4.3)
Multi Events					
All	**7**	**27**	**0.22 (0.10-0.49)**	**49.2 (25.2-74.5)**	**10.9 (2.7-21.4)**
Gastroc			0.07 (0.02-0.30)	42.0 (24.0-60.0)	3.1 (0.0-8.4)
Soleus			0.15 (0.06-0.39)	52.8 (18.5-87.0)	7.8 (0.5-17.5)
Non-specific			-	-	-
Sprints					
All	**90**	**265**	**0.11 (0.07-0.15)**	**28.5 (14.4-48.0)**	**3.0 (1.2-6.3)**
Gastroc			0.04 (0.02-0.07)	32.2 (5.7-75.4)	1.3 (0.1-3.4)
Soleus			0.05 (0.03-0.09)	29.2 (16.3-45.6)	1.5 (0.5-2.8)
Non-specific			0.01 (0.00-0.04)	11.7 (2.0-19.0)	0.1 (0.0-0.3)
Throws					
All	**16**	**45**	**0.04 (0.01-0.18)**	**22.0 (0.0-44.0)**	**1.0 (0.0-2.9)**
Gastroc			-	-	-
Soleus			0.04 (0.01-0.18)	22.0 (0.0-44.0)	1.0 (0.0-2.9)
Non-specific			-	-	-

Note: Injury incidence is presented without adjustment for sex, age, and prior injury

Discussion

CMI incidence, severity and time-loss burden

CMI appears to have a high incidence proportion affecting approximately 11.1% of all athletes involved in this study, with the highest proportion of injuries to the soleus muscle. CMI has a peak incidence in April which can have a detrimental effect on an athlete's performance goals for the

impending summer competition season. Overall, there were more injuries to the right lower limb, with some event groups tending to be more injured on the right leg. Ninety percent of injuries were time loss and 33% of all these injuries were classed as severe. Recurrence of calf muscle injury was 16% with a small cohort of athletes sustaining multiple calf strains over the study period. There appeared to be a relationship between CMI and athlete age or previous injury to the lower leg. There was no relationship between athlete sex and CMI occurrence. Incidence, severity and burden of CMI varied across event groups.

There is limited literature on CMI in elite athletics cohorts, though the literature that is available would suggest soleus is the calf muscle most prone to injury.[7–9] The high incidence of these injuries in athletics could be due to the nature of the events involving some element of running whether that be in the track or in the field. Specifically, the soleus is one of the key muscles in generating muscular power in running, jumping and accelerating. [14–16,21] The specific demands on soleus during all running speeds where it needs to produce 6-8 times body weight could be a possible explanation for a higher incidence of this specific muscle group being injured [14,15]. Nonetheless, the incidence across athletics and other running sports makes a priority to perform further research and inform prevention and rehabilitation practice.

The severity of CMI is a factor that needs to be respected for athletes, coaches and practitioners working in athletics. With one third of the cohort experiencing a severe calf injury, coupled with a mean time loss of 29 days, there could be profound implications to an athlete's performance particularly if the injury occurred in a critical part of the performance season. Green et al (2019) demonstrated overall quicker recovering times for AFL football players, with soleus taking longer to recover in general compared to gastrocnemius. This discrepancy in recovery times in gastrocnemius and soleus has been replicated in this study, though the difference was not significant (27.1 days vs 31.1 days). Within athletics, and other sports, it has been demonstrated that extended time away from training can have an impact on an athlete achieving their physical and performance goals within the same season. [22,23] In this study, peak incidence tended to happen just ahead of the traditional northern

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hemisphere competition season (April). CMI at this point of the season would therefore be very disruptive and a concerted effort to reduce the impact of this specific injury at this unique time point is warranted.

The burden of CMI in this cohort is comparable to other studies and other sports.[5] Kelly et al (2020) demonstrated that soleus and gastrocnemius muscle strain in elite track and field athletes had a burden of 2.2 days lost per athlete year, yet when considered across all events, that study does not differentiate different levels of risk across event groups. In the current study, burden varied widely across event groups and the event groups with the highest burden were hurdles, middle distance and multi event athletes. Of all the event groups, the multi eventers experienced the highest burden with the severity of CMI being the main factor to increasing the burden figure. The reason for high severity and burden in multi eventers may be because of highly varied physiological and biomechanical demands placed on the calf during the multitude of events and training demands these athletes must perform daily.

Recurrence and calf muscle injury

A characteristic unique to this study is the proportion of recurrence and length to time to recur. Compared to other sports, the percentage recurrence was lower and the average time to recurrence was longer. From studies within AFL, soleus was the main muscle to reinjure (91.4%) which was replicated in this study where 100% of reinjuries were to the soleus.[6] The soleus muscle is believed to have a high recurrence rate because of the varied and large physiological and biomechanical demand places on it over a wide range of athletic tasks and running speeds.[14–16,21] The central tendon residing within the soleus muscle may also be a factor to injury recurrence as intratendinous muscle injuries have been demonstrated to occur more frequently in other studies on muscle injury research.[24–27] Recurrences are typically associated with longer return to play times, this was not replicated in this study where CMI recurrence were very close to index injury recovery times (index injuries 33.2 vs recurrences 29.7 days).[6,24] The difference with other sports may be explained by the recurrences in athletics occurring almost one year after the index injury, ensuring complete physiological muscle,

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aponeurotic and tendon tissue healing occurring in this time frame.[28–30] It is difficult conclude what factors explain the difference in both recurrence rates and time to recurrence compared to other sports though further research is warranted

Age and previous lower limb injury associated with calf muscle injury

It is well documented that age is a risk factor for muscle injury of the hamstrings, quadriceps, adductors and the calves[10,31–33] Research in AFL football over several years establishes age as a risk factor for CMI.[31] Similarly, Hagglund et al (2013) found that the risk of CMI doubled within older players in the most elite European football leagues.[32] Green et al (2017) were able to conclude that age is a risk factor for CMI in a systematic review of calf strain in a mixed group of athletes.[10] The findings of the current study would contribute to the body of evidence related to CMI injury risk as there was a positive association between older athletes and occurrence of CMI. It is thought that age related changes in skeletal muscle morphological and neurological characteristics have been suggested as the reason for increased risk of muscle injury.[34,35] Specific qualities such as a decline in rate of force development and absolute power are modifiable, and therefore, may provide an avenue for assessment, monitoring and conditioning of the older athlete in relation to developing injury prevention and performance programs.

Prior injury to the same calf muscle, and also injury to the contralateral limb, have been demonstrated to be risk factors to CMI in athletes from multiple sports.[10] There is a similar injury pattern in athletics where calf strain on any lower limb region puts you at further risk of another strain. Because of the role of the calf as a force attenuator, stress shielding other injured body sites could be an explanation as to why contralateral muscle injury appears to be a risk to future injury in athletics and other sports.

Event group analysis and calf muscle injury

CMI analysis by event group has not been done before in track and field. The findings of this study indicate that athletes from any event group can be at risk of sustaining CMI. Athletes who participated in jumps, sprints, multi-events and middle distance tended to have more injuries on the right side.

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Limb dominance for an athlete's take off leg, or, the demands of bend running could be possible explanation for the right lower limb being more effected by CMI. Bend running on a circular track has been shown to change the functional demands of the right-side lower limb by altering kinematics around the knee and hip in maximum velocity running.[36] Running on a curved track repeatedly is known to change muscle architecture and EMG patterns particularly of the soleus and gastrocnemius muscle responses to those biomechanical demands.[37] Peak force production has been demonstrated to be higher when running at 6m/s, or maximal velocity sprinting, which has implications for middle distance and sprinting athletes who run bends consistently as part of their training and competiton.[38] Though there has been some research on bend running biomechanics, none has been utilised to establish a relationship to CMI injury risk, therefore further analysis of bend running and its relationship to injury could be an avenue for future research.

Limitations

There were several limitations to the conduct of this study. As recommended by the IOC consensus on injury surveillance the units for exposure in this cohort with differing athletic demands is 'per athlete year'. Typically, exposure would be expressed as per 1000 hrs for training/competition but within this cohort monitoring of this nature is not feasible given the geographical spread of all athletes on the program and heterogenous nature of athletic events. Another limitation is the omission of specific muscle injury classification, i.e. BAMIC grading, within this paper. This paper's focus is on CMI epidemiology and standardised radiological re-examination of every CMI was not performed for practicality and applicability to clinical practice. This study's analysis would have been enriched with deeper statistical analysis of between group comparisons but at statistical power was not high enough due relatively small number of injuries analysis of this nature was not possible.

Conclusion

CMI is common in elite track and field athletics though the incidence, severity and burden of this injury varies across event groups. CMI has a peak incidence in April which can have a detrimental effect on an athlete's performance goals for the impending summer competition season, therefore,

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prevention efforts could be increased in anticipation of this stage of season. Age and previous injury appear to be risk factors associated with CMI incidence. Recurrence rates of CMI in the elite athletic population are relatively low, but unique to the soleus. Bend running and limb dominance may be a factor increasing risk of CMI in this population. Due to the burden and potential performance impact of CMI, further injury prevention efforts are required to reduce the impact of this injury type.

Practical Implications:

- CMI is common in elite track and field athletes with all event groups being at risk of injury.
- The soleus muscle is the most injured muscle with the biggest burden and recurrence making it a muscle of focus when developing rehabilitation and prevention programs.
- CMI occurred most frequently at the start of the athletics competitive season which could negatively impact on an athlete's chances of success that season due to the average severity of these injuries.
- Older athletes, and athletes with a previous lower limb injury, are more at risk of CMI in elite athletics
- Hurdlers, multi eventers and middle-distance runners were most injured potentially indicating that the high and diverse demands of athletic performance in these events could be a risk factor for injury

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Declaration of Generative AI and AI-assisted technologies in the writing process

There has been no use of generative AI technology at any stage of the research.

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Declaration of interests

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐ The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

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Ethical Statement

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